NVIDIA Isaac for Accelerated Robotics

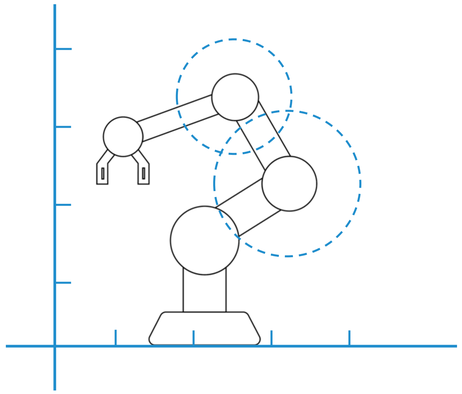
Workshop Agenda

9:00am	Welcome	• Setting up • Introductions and workshop overview
9:15am	OpenUSD for Robotics	• Building blocks of robots in OpenUSD • Importing robots to OpenUSD • Assembling robots and environments
11:00am	Software-in-the-Loop Testing for Robotics	• Integrating ROS with Isaac Sim • Configuring ROS graphs for robots • Creating occupancy maps • Autonomous navigation with Nav2 • Manipulation with MoveIt2
1:00pm	Lunch	
1:30pm	Accelerating ROS 2 Workloads	• Overview of Isaac ROS • Grasp authoring • Pick and place simulation
2:45pm	Synthetic Data Generation	• Overview of physical AI and simulation • Collecting data with GR00T-Teleop • Augmenting data with GR00T-Mimic • Training robots using Isaac Lab • Overview of Cosmos World Foundation Models (WFMs)
4:30pm	Review and Assessment	• Discussion and review

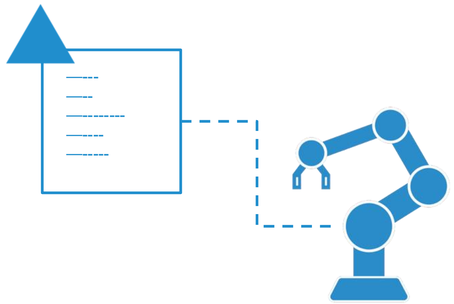
NVIDIA Isaac for Accelerated Robotics, Part 1

OpenUSD for Robotics

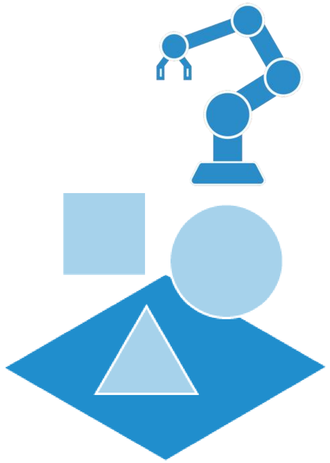
Workshop Objectives



Learn the building blocks of a robot in USD

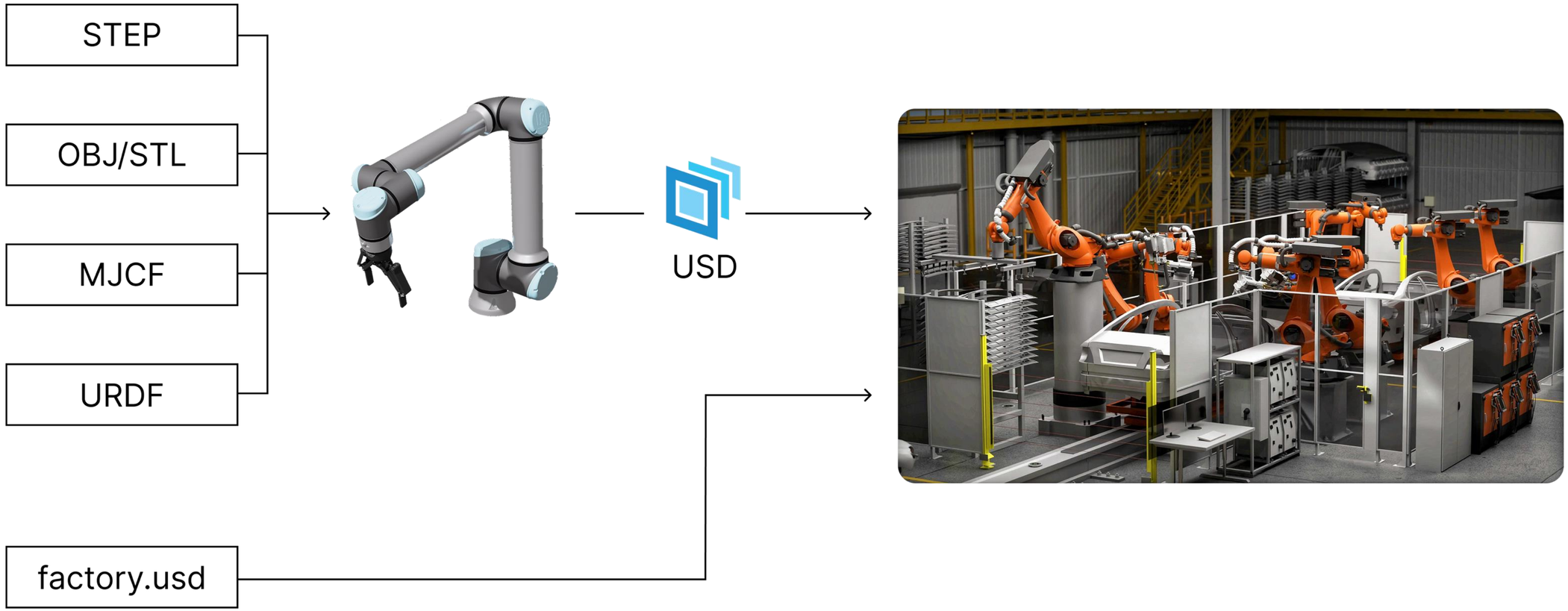


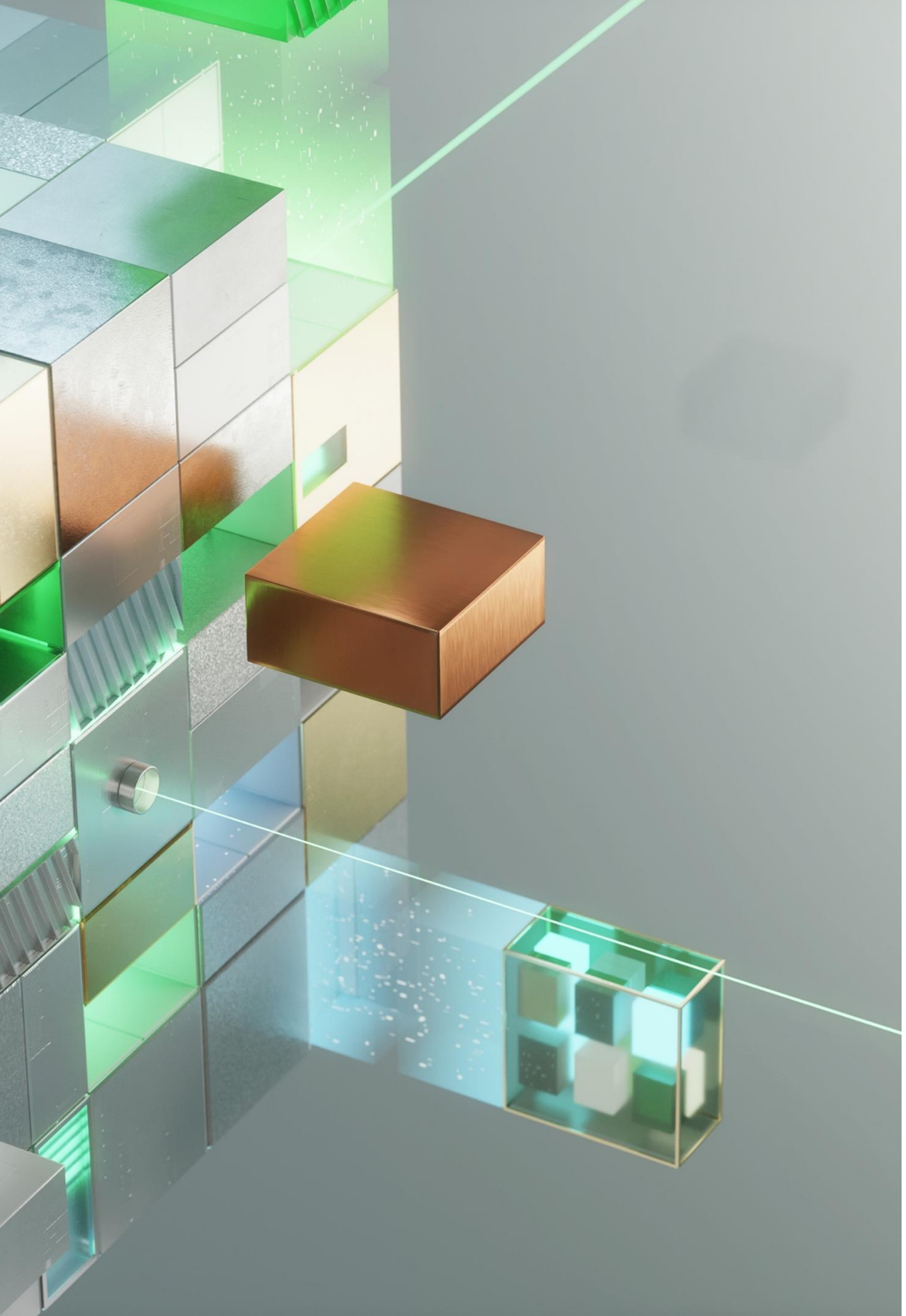
How do I import robots to USD



How can I assemble robots and Environments in Isaac Sim

Anatomy of a Robot

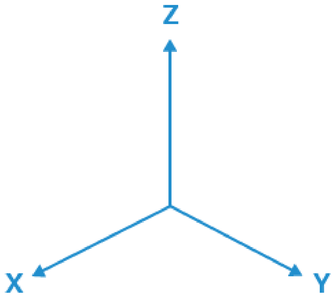


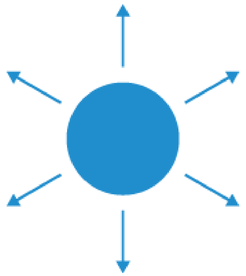


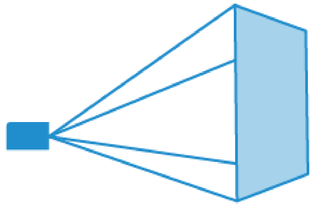
Universal Scene Description (USD) Overview

USD Overview

Prim



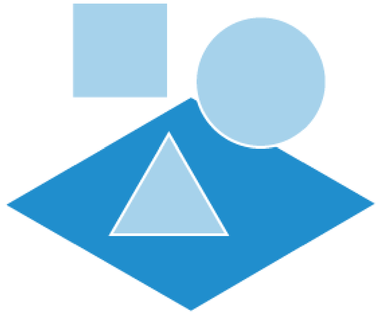




A **Prim** is the primary container objects in USD

USD Overview

Prim

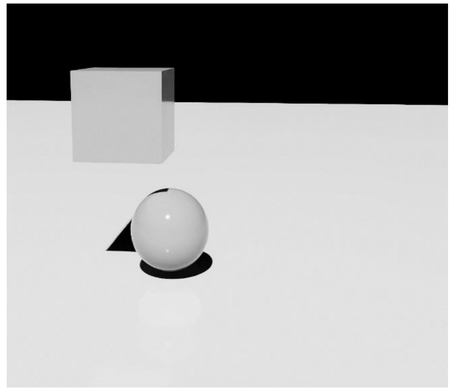


Geometry

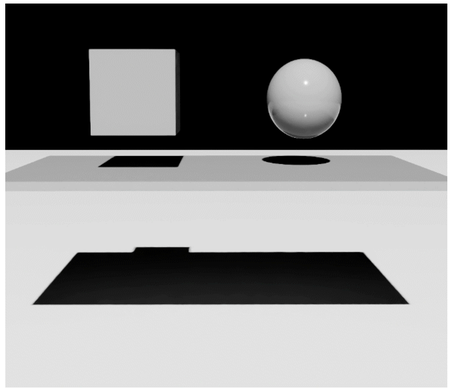
Geometry is a simple, pre-defined shape such is cube or cylinder

USD Overview

Robot ingredients



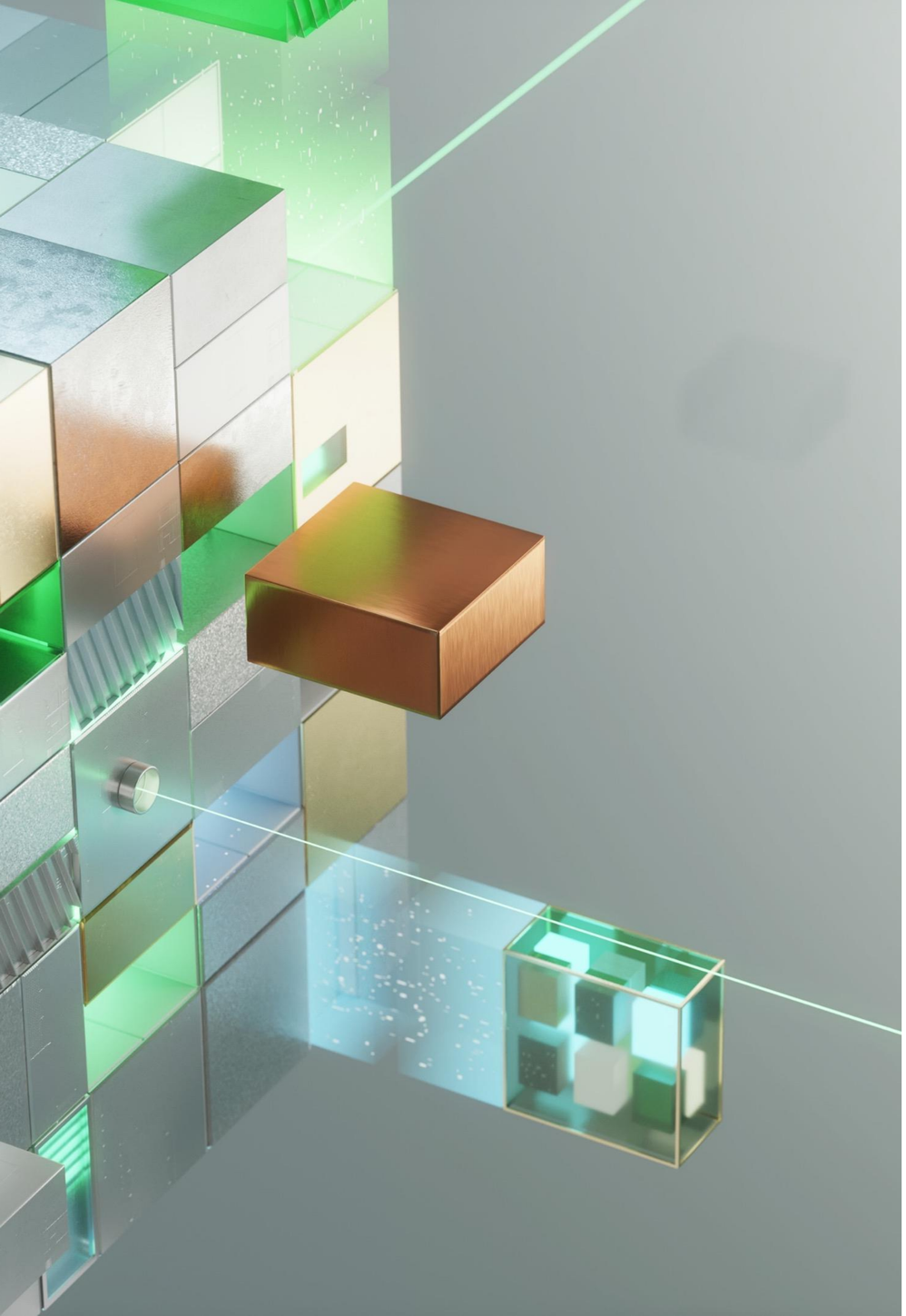
Rigid body



Colliders

Joints

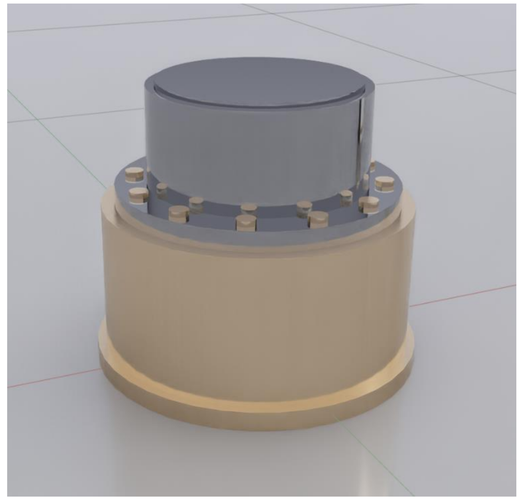
Articulation



Joint Types

Joint Types in USD

And their URDF counterparts



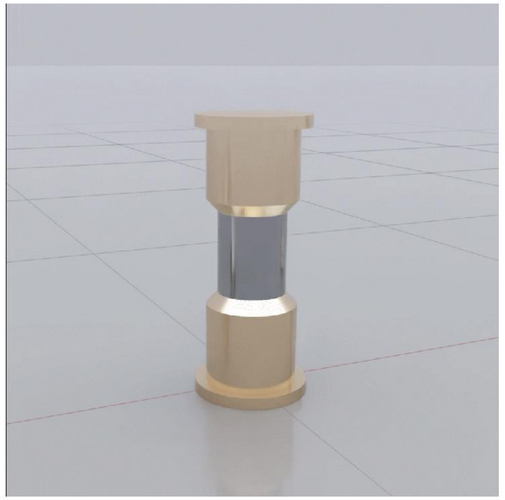
Fixed

No motion

Revolute

In URDF can also be continuous

Angular motion
with/without limits



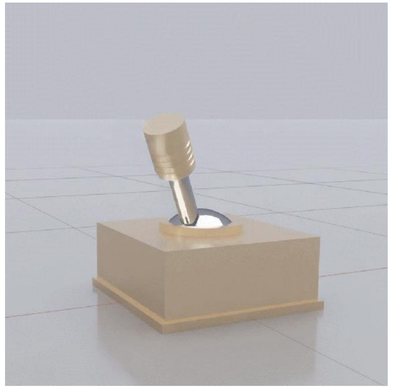
Prismatic

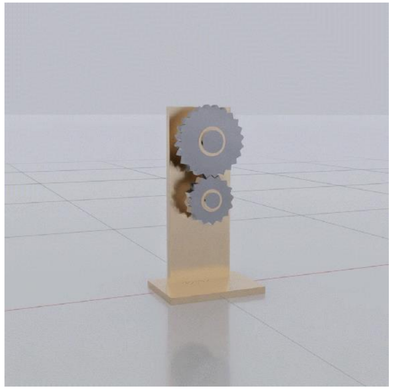
Linear motion
with/without limits

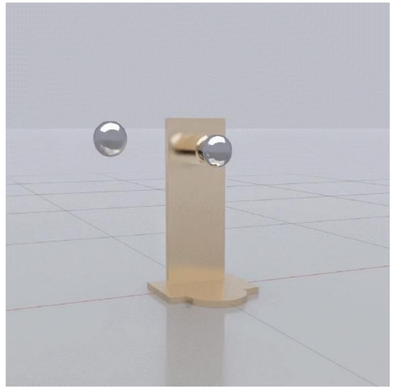
Joint Types in USD

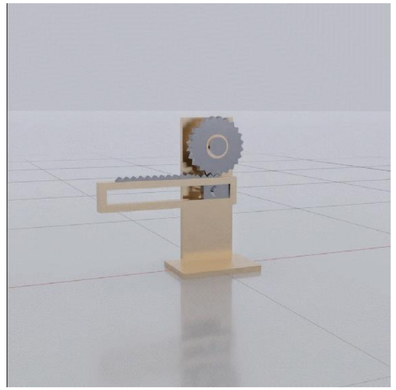
And their URDF counterparts

USD Only - These joint types don't exist in URDF









Spherical

Allow motion in X, Y, Z rotation
Equivalent to 3 revolute joints

D6

Allow motion in selected DOF
if all selected, it is equivalent to floating joint

Gear

Define relation between two **revolute joints** with a **ratio**

Distance

Total freedom between two bodies - until distance reached

Rack and Pinion

Define relation between **revolute and prismatic joint**

Joint Types in USD

And their URDF counterparts

Mimic

Attribute, not a joint

Define relation between
two **joints** with a **gear ratio**

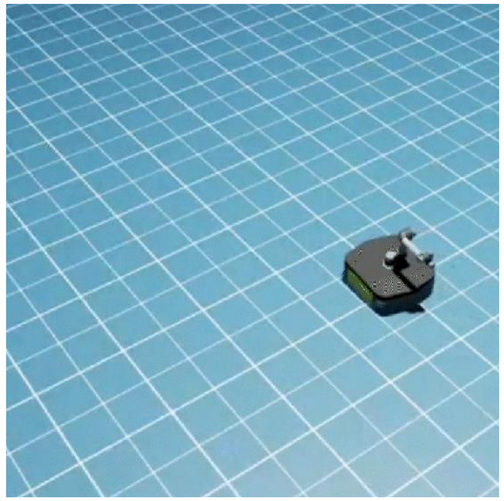
Joint Types in USD

And their URDF counterparts

URDF Only - These joint types don't exist in USD

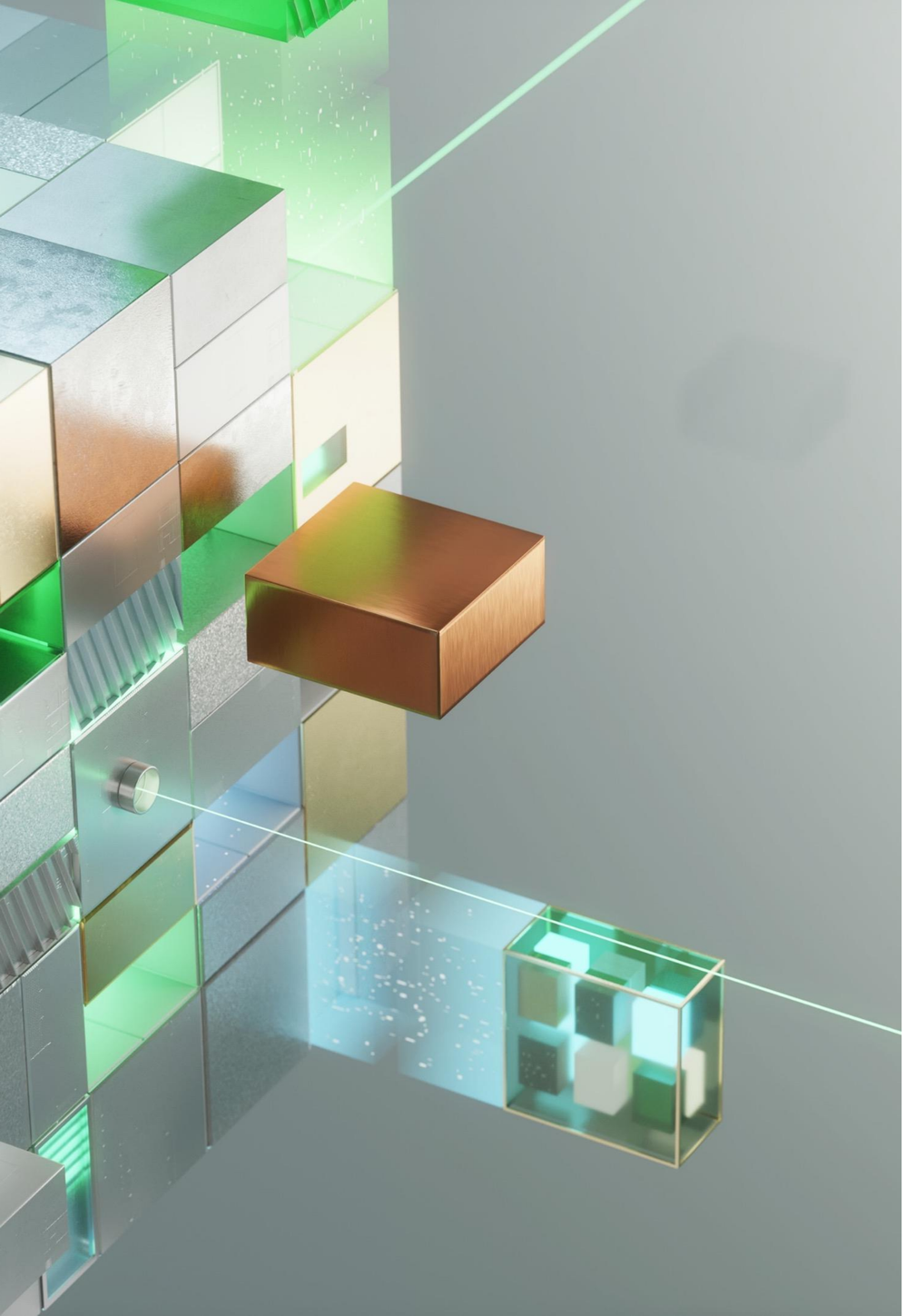
Floating

Total freedom in all DOF



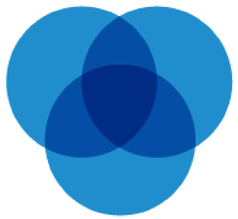
Planer

Total freedom along a plane

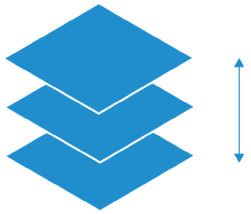


USD Stage Composition

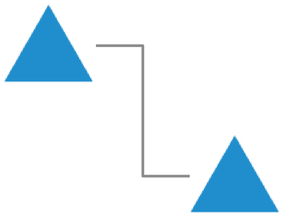
USD Stage Elements and Composition



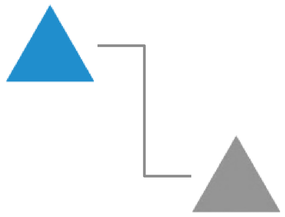
Materials



Layers

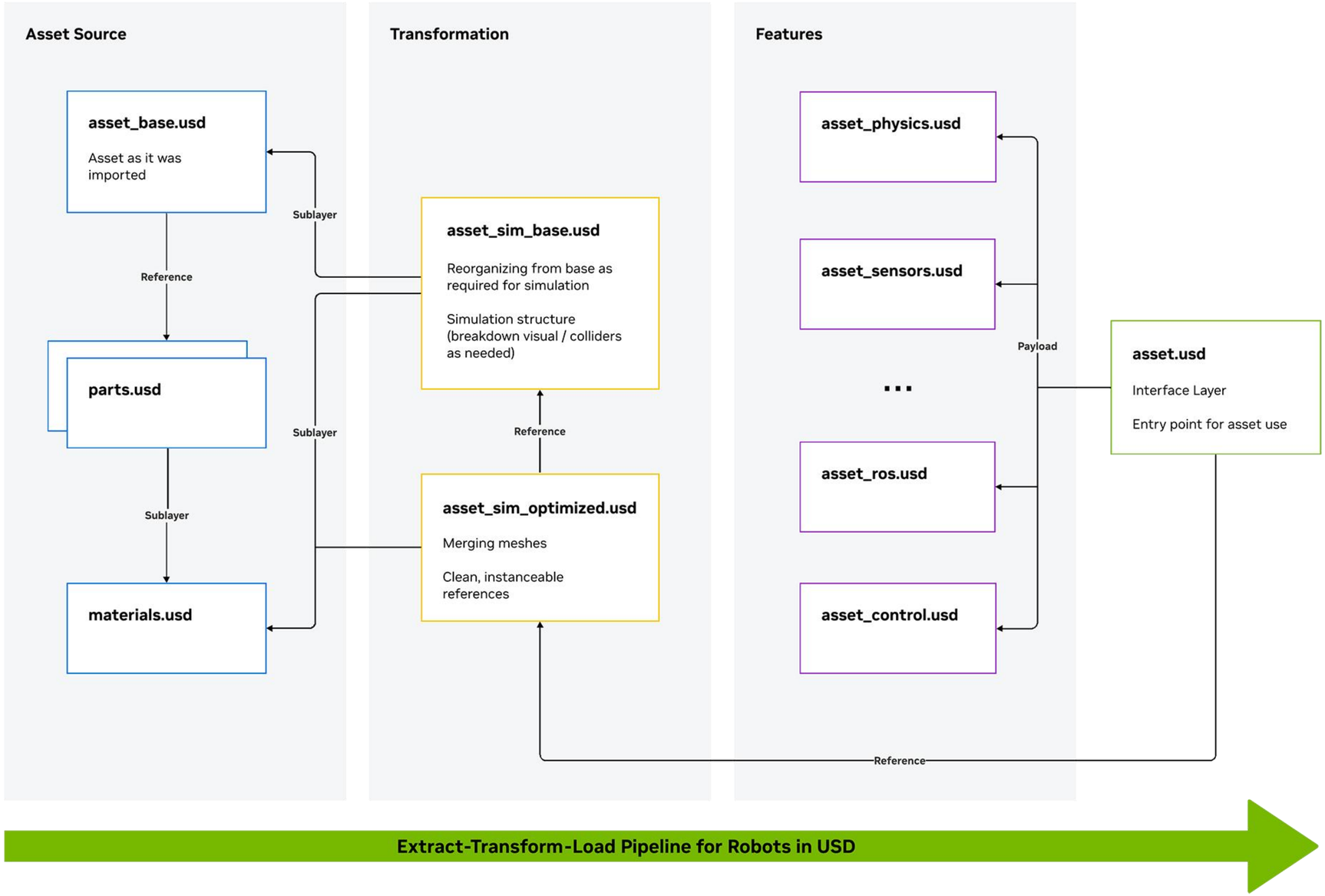


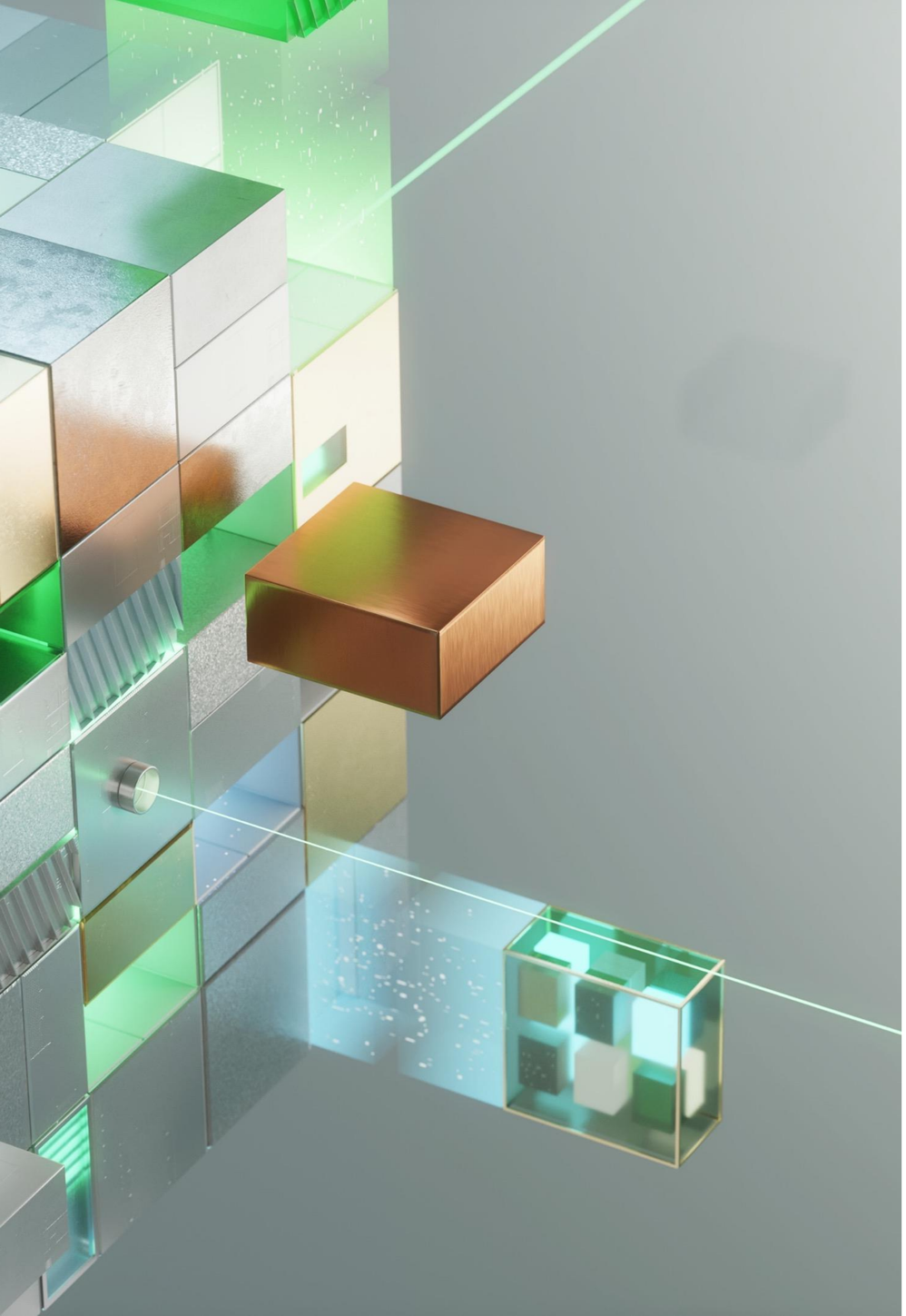
References



Overrides

USD Stage Elements and Composition

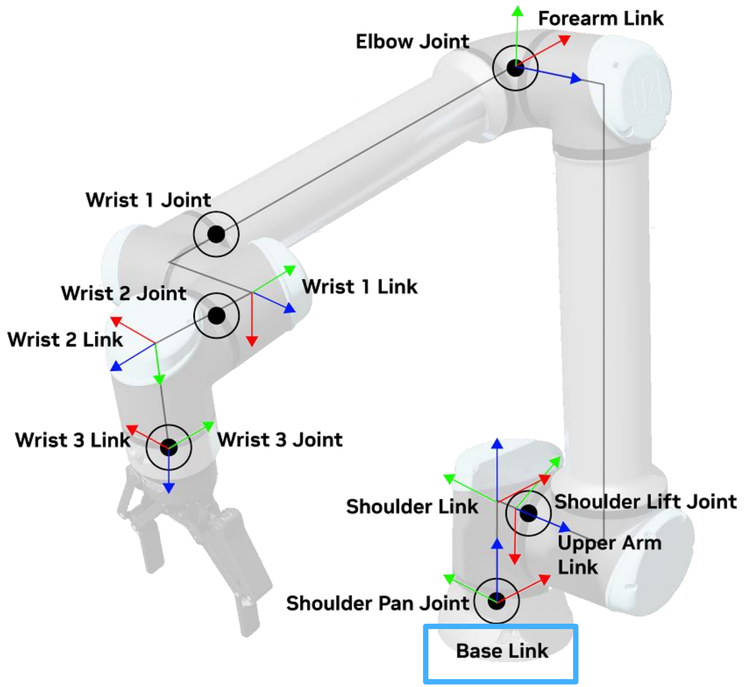




Hands-On:
Assemble UR10e Robotics Arm and 2F-140 Gripper

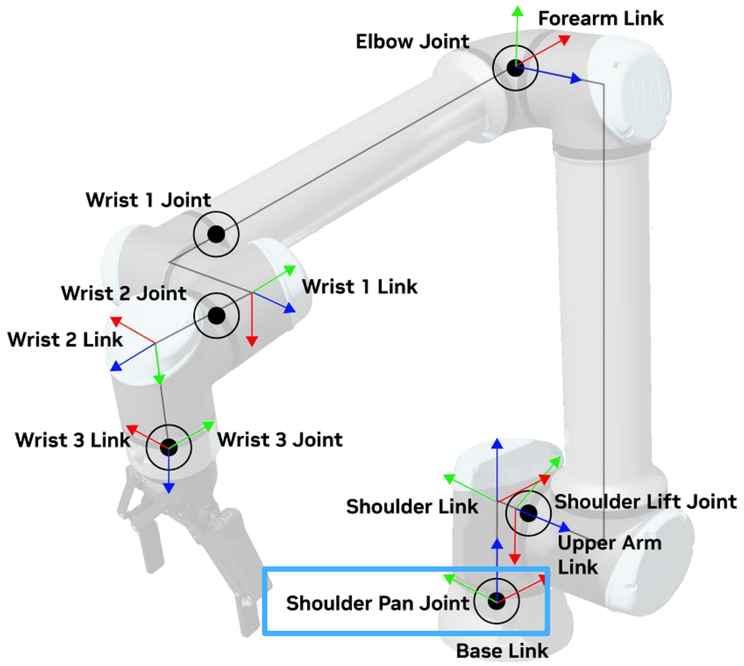
Structure of a URDF

```
  <link name="base_link_inertia">
    <visual>
      <origin rpy="0 0 3.141592653589793" xyz="0 0 0"/>
      <geometry>
        <mesh filename="package://ur_description/meshes/ur10e/visual/base.dae"/>
      </geometry>
      <material name="LightGrey">
        <color rgba="0.7 0.7 0.7 1.0"/>
      </material>
    </visual>
    <collision>
      <origin rpy="0 0 3.141592653589793" xyz="0 0 0"/>
      <geometry>
        <mesh filename="package://ur_description/meshes/ur10e/collision/base.stl"/>
      </geometry>
    </collision>
    <inertial>
      <mass value="4.0"/>
      <origin rpy="0 0 0" xyz="0 0 0"/>
      <inertia ixx="0.0061063308908" ixy="0.0" ixz="0.0" iyy="0.0061063308908" iyz="0.0"
izz="0.01125"/>
    </inertial>
  </link>
…
```

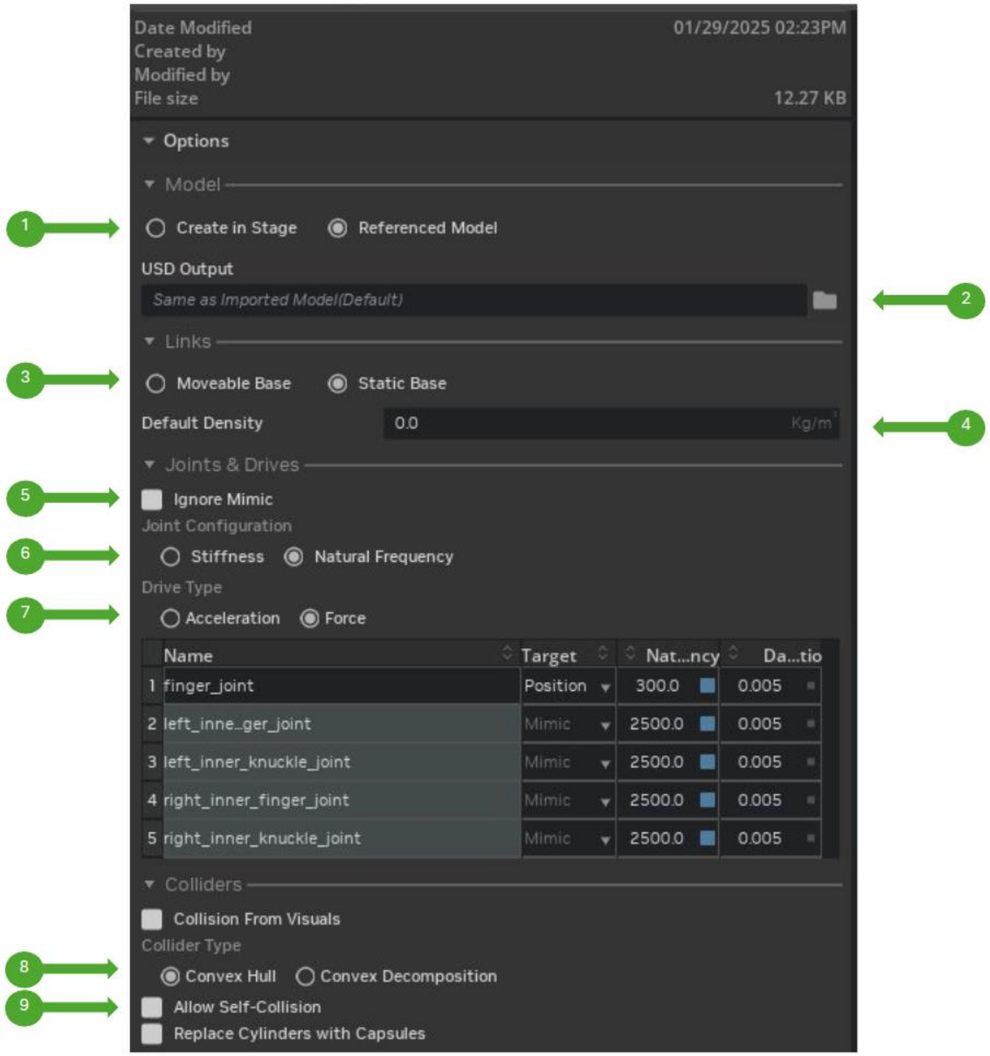


Structure of a URDF

```
  <joint name="shoulder_pan_joint" type="revolute">
    <parent link="base_link_inertia"/>
    <child link="shoulder_link"/>
    <origin rpy="0 0 0" xyz="0 0 0.1807"/>
    <axis xyz="0 0 1"/>
    <limit effort="330.0" lower="-6.283185307179586" upper="6.283185307179586"
velocity="2.0943951023931953"/>
    <dynamics damping="0" friction="0"/>
  </joint>
…
```

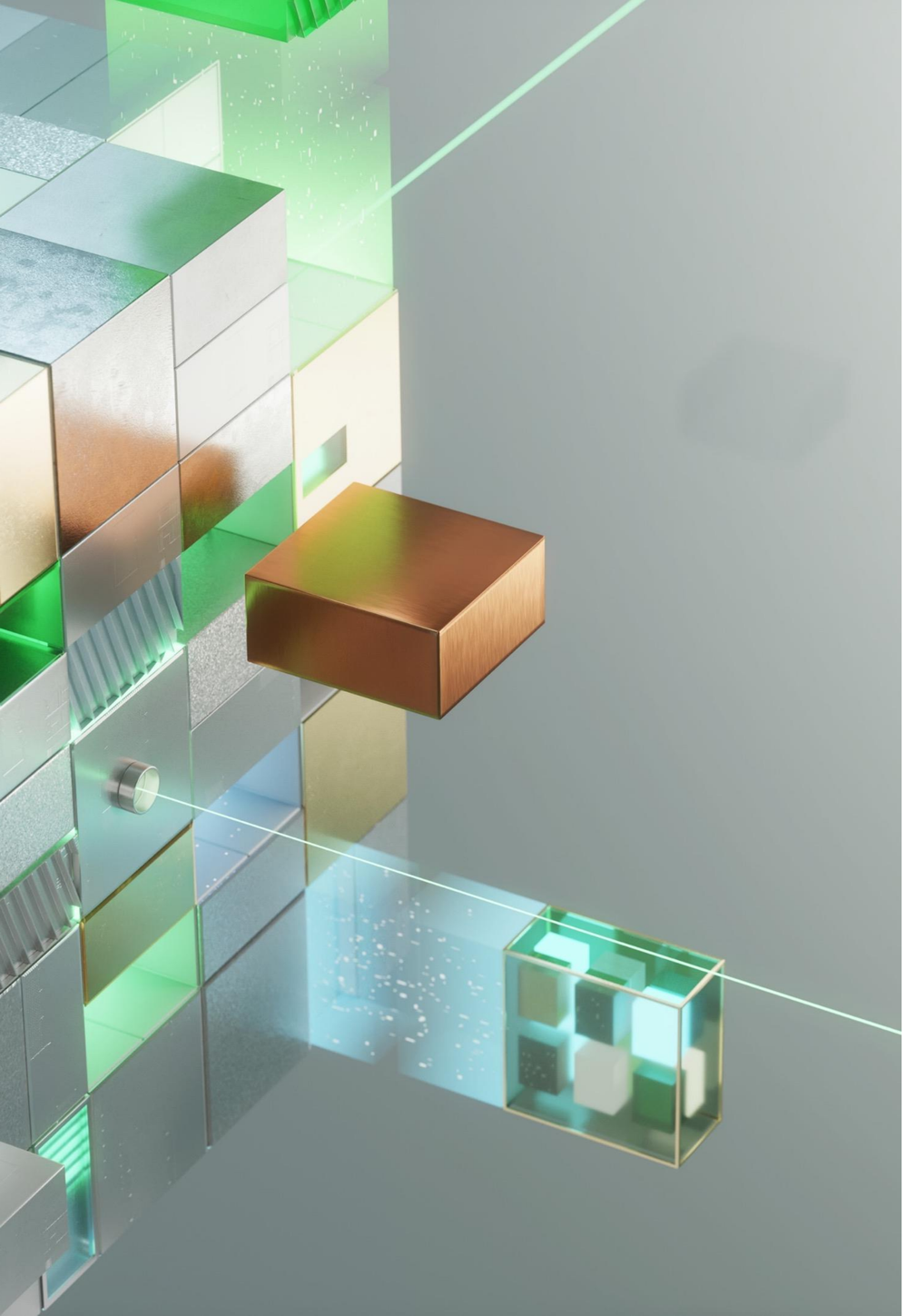


URDF Importer Graphical User Interface



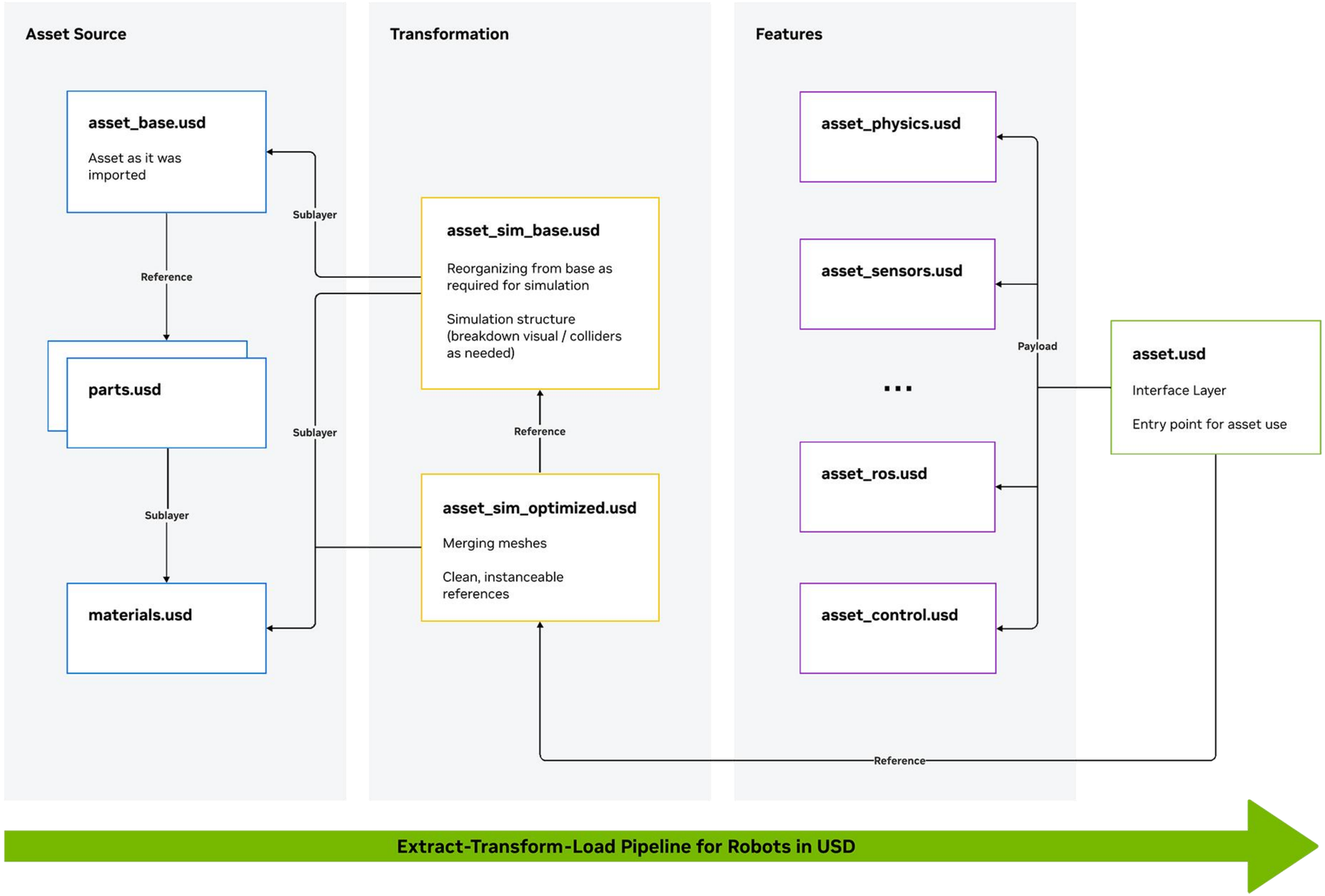
Reference	Function
1	Determine to spawn the asset in the current opened stage or link as a reference to the stage
2	Output file path of the USD
3	A fixed joint will be created between the robot base link and world if “Static Base” is selected
4	If set to 0, it will use the mass specified in the URDF or 1000/m^3 (water density) if no mass is specified
5	Ignore mimic joints
6	Joint configuration, to either directly set stiffness and damping or compute them indirectly natural frequency
7	Joint settings, force will account for the mass and inertia of the body link
8	Collider configurations if collider shapes are not present in the URDF
9	Allow Self-Collision will enable the robot to collide with itself

Get to work!

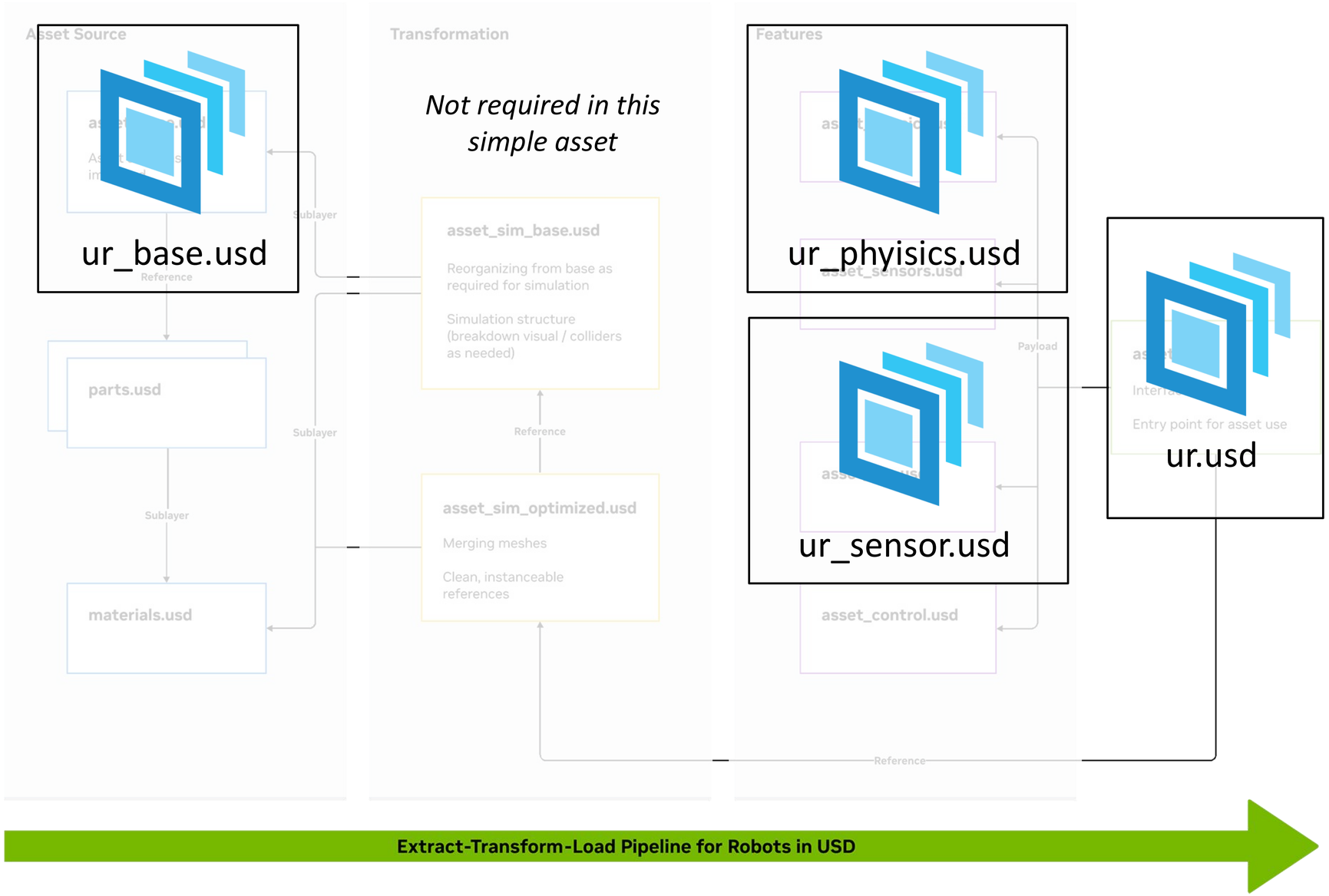


Case Study:
URDF Importer Asset Structure

URDF Importer Asset Structure

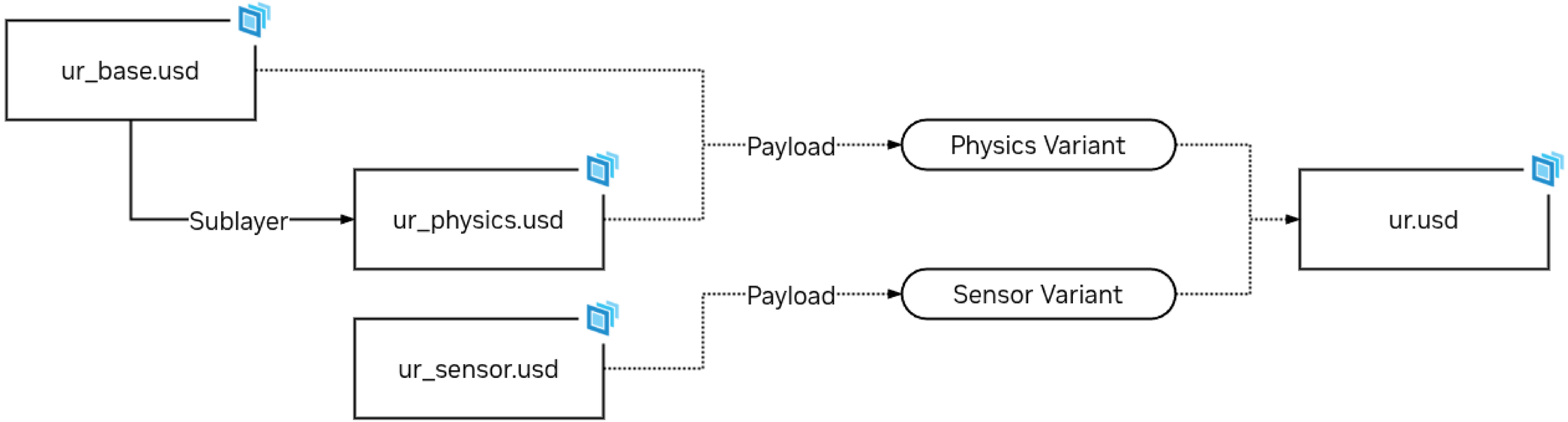


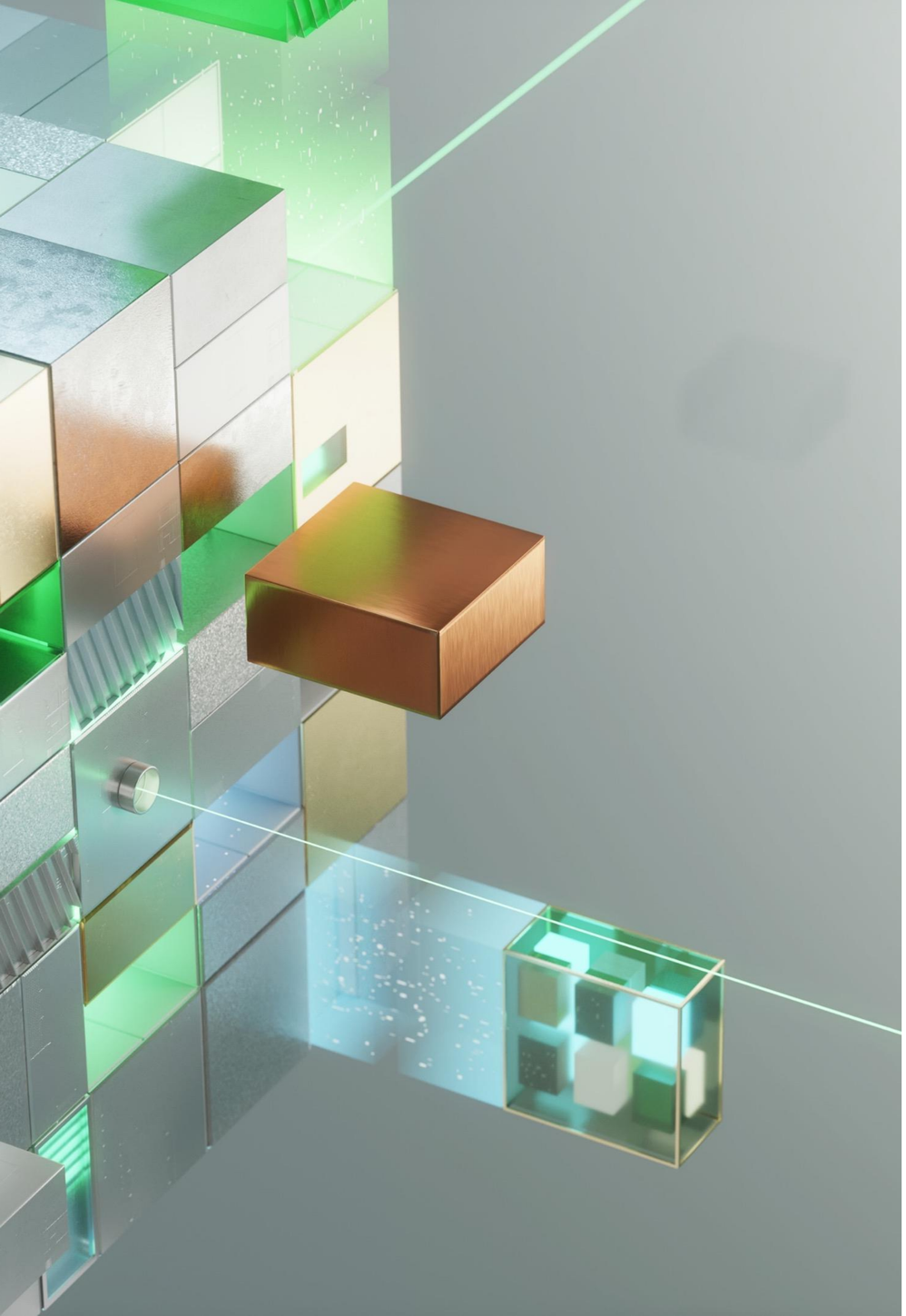
URDF Importer Asset Structure



URDF Importer Asset Structure

Composition of the robot assets

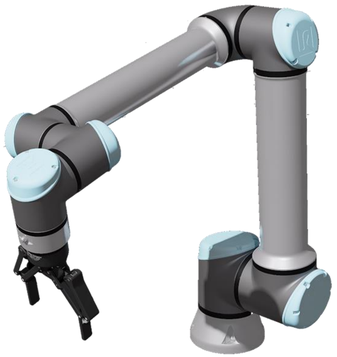




Hands-On:
Assemble a Robot Training Scenario

URDF Importer Asset Structure

Composition of the robot assets



ur.usd

Interface Layer

Entry point for asset use

reference

sortbot.usd

reference

small_warehouse...

reference

training_scene.usd

Interface Layer

Entry point for asset use

NVIDIA Isaac for Accelerated Robotics, Part 2

Software-in-the-Loop Testing

(Hands-On Only)